

GEOSPATIAL AI · APPLIED RESEARCH

Map Scientist

Geospatial AI · Map Inference · Road Intelligence · Research-to-Production

 Tasco Building, Hanoi – Pham Hung

 TASCO Mapping

 Individual Contributor

 Reports to Head of Mapping

ABOUT THE COMPANY

About Tasco

VISION

A technology-enabled investment ecosystem, integrating Smart Mobility, Insurance, and Investments in essential services and innovation.

STRATEGIC PARTNERS

Geely Auto

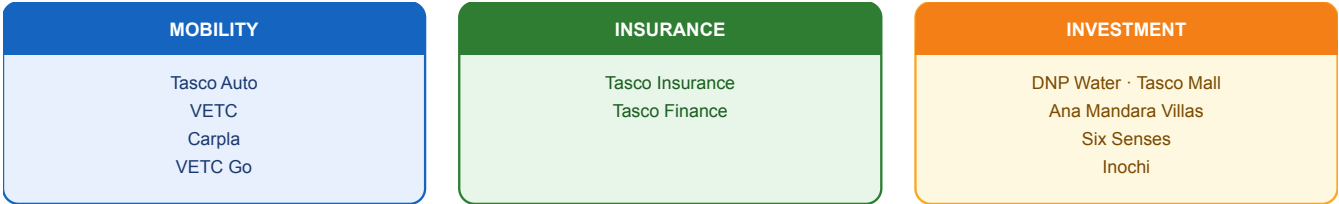
Mitsui & Co.

IFC – World Bank Group

Corporate Information

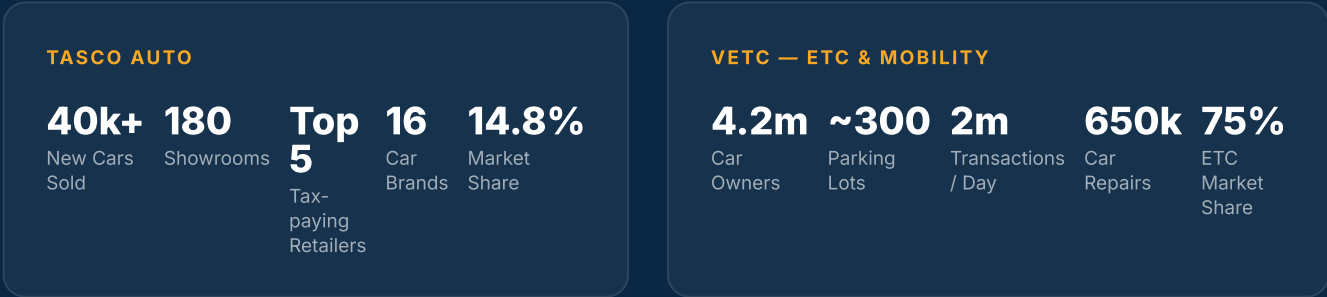
Stock Ticker	HUT (HNX)
Established	1971
Subsidiaries & Affiliates	88
Employees	~10,000
Consolidated Sales	\$1.4 billion
Total Assets	\$1.98 billion
Website	tasco.com.vn

BUSINESS ECOSYSTEM



WHY TASCO

Our Mobility Ecosystem in Numbers



0 Job Overview

POSITION	Map Scientist
FOCUS	Geospatial AI, Road Attribute Inference, Lane-Level Map Intelligence, Map Quality Science
DEPARTMENT	TASCO.COM.VN Mapping
LOCATION	Tasco Building Hanoi, Pham Hung
LEVEL	Individual Contributor
REPORTS TO	Head of Mapping

Job Purpose

The Map Scientist supports the development of AI-assisted map-making systems for Tasco Maps, MyTasco, VETC mobility services, navigation, HD Maps / ADAS Maps, and future vehicle intelligence products.

This role sits between Map Operations, Data Science, AI Engineering, and Product — translating mapping problems into measurable scientific tasks: dataset design, annotation schema, model benchmarking, geospatial feature engineering, error analysis, and prototype development.

It is an **applied research role** focused on turning aerial imagery, street-level imagery, GPS probes, OSM / Overture data, traffic-sign data, POI data, and user feedback into scalable map intelligence for Vietnam.

HOW THE ROLE WORKS — DATA TO MAP INTELLIGENCE



A. Geospatial AI Research & Prototyping

- Read, summarize, and reproduce selected papers related to road graph extraction, lane-level topology, road attribute inference, map-matching, traffic-sign detection, POI extraction, and map quality evaluation.
- Build small-scale prototypes for map inference tasks using Python, PostGIS, PyTorch, computer vision, graph algorithms, and geospatial data processing.
- Translate research ideas into production-oriented experiments with clear inputs, outputs, baselines, metrics, limitations, and rollout risks.
- Maintain internal research notes, experiment logs, and technical memos for the Mapping team.

B. Road Intelligence & Lane-Level Map Inference

- Support development of models and rules to infer road attributes from aerial imagery, street imagery, OSM, GPS probes, and field data.
- Work on lane count estimation, road width inference, turn-lane detection, road-surface classification, intersection structure analysis, and traffic-signal / traffic-sign context extraction.
- Assist in building lane-level and intersection-level datasets for Hanoi, Ho Chi Minh City, expressways, toll stations, parking zones, service roads, and high-priority corridors.
- Evaluate model outputs against ground truth from MapOps, field collection, authoritative data, and manual QA.

C. Map Quality Science & Confidence Scoring

- Design quantitative metrics for map data quality: completeness, freshness, positional accuracy, topology correctness, attribute accuracy, false-positive rate, false-negative rate, and operational impact.
- Build confidence scores for map objects such as road segments, turn restrictions, lane attributes, traffic signs, POIs, entrances, geofences, and addresses.
- Analyze disagreement between multiple data sources: OSM, Overture Maps, satellite imagery, street imagery, GPS traces, government data, user reports, and internal business data.
- Prioritize MapOps work using risk-based scoring: high-traffic roads, high-fine-risk areas, navigation-critical intersections, and stale data regions.

D. Data Annotation, Benchmarking & Evaluation

- Define annotation guidelines for AI map tasks: lane marking, stop line, turn arrow, traffic sign, road boundary, POI signboard, entrance, parking polygon, toll station, and road access point.
- Build benchmark datasets with clear train / validation / test splits by city, road class, imagery source, and complexity level.
- Establish baseline models and compare performance using precision, recall, F1, IoU, topology correctness, graph connectivity error, routing impact, and human-review cost.
- Conduct systematic error analysis to identify whether errors come from data quality, annotation inconsistency, model design, imagery quality, road occlusion, GPS noise, or ambiguous legal rules.

E. AI-Assisted Map Operations

- Build tools that help MapOps review model predictions faster: candidate generation, object ranking, auto-flagging, active learning queues, and QA dashboards.
- Work with MapOps Specialists to convert AI predictions into validated map updates.
- Create SOPs for human-in-the-loop validation of AI-generated map features.
- Support a practical pipeline where AI proposes, MapOps verifies, Engineering publishes, and Product monitors user impact.

F. Data Sources, Fusion & Validation

- Integrate and validate signals from OSM, Overture Maps, satellite imagery, street-level imagery, GPS probes, government data, field collection, user reports, and Tasco/VETC business data.
- Design conflation logic to reconcile geometry, topology, attributes, source reliability, and temporal freshness across datasets.
- Support source reliability scoring, data lineage, rollback evidence, and review workflows for critical map features.
- Partner with Engineering to expose validated outputs to internal map editor tools, QA dashboards, search systems, routing graph generation, and release pipelines.

G. Documentation, SOPs & Team Enablement

- Build documentation for dataset specs, annotation rules, benchmark design, model evaluation, confidence scoring, and release readiness.
- Train junior MapOps members and contributors on AI-assisted validation and map quality standards.
- Work with Product, Engineering, Data Science, and Business teams to shape the map intelligence roadmap.

MUST-HAVE

- ✓ Bachelor's degree in GIS, Geomatics, CS, Data Science, EE, Transportation Engineering, Applied Mathematics, or related field.
- ✓ 0–2 years in geospatial data, ML, computer vision, map operations, remote sensing, navigation, logistics, or mobility data products.
- ✓ Strong Python skills for data processing, geospatial analysis, experiment scripting, and automation.
- ✓ Working knowledge of SQL and PostgreSQL/PostGIS.
- ✓ Familiarity with geospatial data formats: GeoJSON, Shapefile, GeoPackage, PBF, Parquet, raster tiles, vector tiles, GeoTIFF.
- ✓ Basic understanding of OpenStreetMap: nodes, ways, relations, tags, turn restrictions, road classes, POIs, admin boundaries.
- ✓ Basic ML understanding: supervised learning, model evaluation, train/test split, overfitting, precision/recall, confusion matrix, error analysis.
- ✓ Strong written communication in English.

NICE-TO-HAVE

- ✓ Experience with PyTorch, TensorFlow, scikit-learn, XGBoost, CatBoost, or similar ML frameworks.
- ✓ Computer vision: YOLO, U-Net, SegFormer, Mask2Former, SAM, OCR, or object detection / segmentation pipelines.
- ✓ Graph algorithms, road graphs, routing engines, map-matching, Valhalla, OSRM, GraphHopper, or NetworkX.
- ✓ Remote sensing, satellite imagery, aerial imagery, street-level imagery, dashcam data, or LiDAR point clouds.
- ✓ QGIS, JOSM, iD Editor, Overpass Turbo, osmium, osm2pgsql, DuckDB, GDAL, rasterio, or GeoPandas.
- ✓ Active learning, human-in-the-loop labeling, data labeling QA, or model-assisted annotation.
- ✓ OSM contribution history, GitHub portfolio, Kaggle / research project, or thesis in mapping, CV, GIS, traffic, or mobility.

SOFT SKILLS & MINDSET

- ✓ Scientific mindset: skeptical, evidence-based, comfortable with uncertainty.
- ✓ Strong attention to data quality and edge cases.
- ✓ Able to turn ambiguous mapping problems into measurable experiments.
- ✓ Comfortable with messy real-world data: noisy GPS, occluded imagery, inconsistent labels.
- ✓ Practical orientation: reducing human review cost and improving product impact.
- ✓ Able to work independently, document assumptions, and ask precise technical questions.
- ✓ Interested in building Vietnam-localized map intelligence rather than generic global models.

4 Success Metrics — 6 to 12 Months

- ♦ Reproduce and document at least 2 relevant mapping / geospatial AI papers or open-source methods.

- ♦ Deliver at least 1 internal benchmark dataset for a priority problem (lane count, turn lane, traffic sign, POI signboard OCR, road width, or map-change detection).

- ♦ Build at least 1 working prototype that generates candidate map updates for MapOps review.

- ♦ Reduce manual review workload for one selected MapOps workflow through AI candidate ranking or active learning.

- ♦ Establish baseline metrics and error taxonomy for at least one map intelligence task.

- ♦ Create confidence scoring logic for at least one map object type: road attribute, POI, traffic sign, entrance, or geofence.

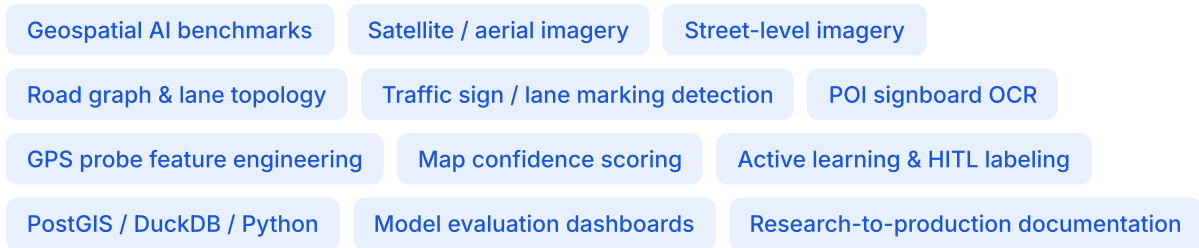
- ♦ Publish internal documentation: dataset spec, annotation guideline, model benchmark, error analysis, and next-step recommendation.

- ♦ Support Engineering / MapOps in moving one prototype toward a repeatable pipeline.

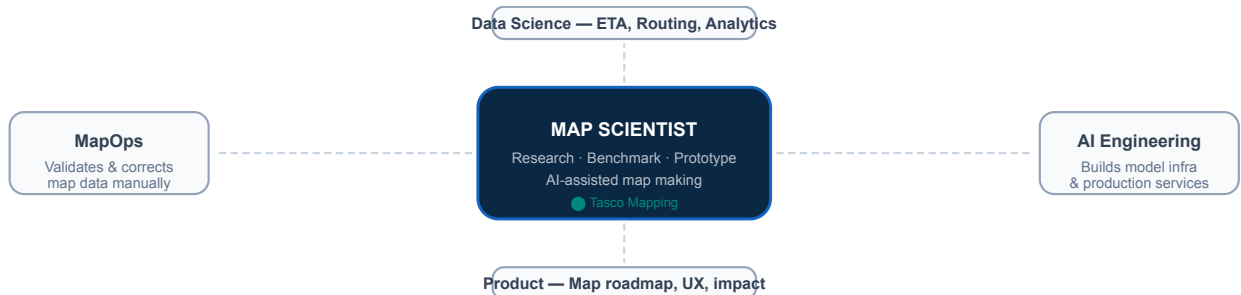
5 Product Scope

Product Line	Responsibility
V-Base-Maps	Road attributes, POI intelligence, map freshness, data completeness
V-Navigation-Maps	Lane guidance, turn restrictions, speed limits, road graph quality
HD Maps / ADAS Maps	Lane-level attributes, road geometry, intersection intelligence
Location & AI Search	POI confidence, signboard OCR, address / entrance signals
Contributor Platform	AI-assisted candidate generation, validation queues, QA scoring
Mobility Data Platform	GPS probe analysis, trajectory features, traffic and safety signals
Map Editor / Internal Tools	Human-in-the-loop review, model-assisted editing, QA dashboard

6 Technical Working Areas



7 How This Role Differs



Role	Main Ownership
Map Operations Specialist	Maintains and validates map data manually / operationally
POI / Geocoding Ops Specialist	Owns POI, address, search, geocoding, and geofence quality
Data Scientist	Builds analytics / ML models for ETA, routing quality, trajectory intelligence, and optimization
AI Engineer	Builds scalable model training and inference infrastructure
Map Scientist	Converts map problems into scientific experiments, benchmarks, prototypes, and AI-assisted map-making workflows

*MapOps ensures the map is correct today. Engineering ensures the system scales. Data Science models trip and route behavior. **Map Scientist** builds the scientific bridge that helps Tasco Maps infer, validate, and update map data faster and more intelligently.*

How to Apply

Send your email with subject **[Tasco Mapping Application - Map Scientist]** with your CV and a portfolio or description of relevant projects in GIS, mapping, computer vision, machine learning, remote sensing, road networks, OSM, traffic, mobility, or geospatial data science.

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[Tasco Mapping Application – Map Scientist]

GitHub repositories, OSM contribution links, notebooks, thesis work, research summaries, benchmark reports, or sample geospatial analysis projects are encouraged.